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| | FINALTERM EXAMINATION<br>FALL 2006<br>MTH601 - OPERATIONS RESEARCH | Marks: 65<br>Time: 120min |
|---|---|---|

StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date:

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6. ***In order to get full marks do all necessary steps.***

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| For Teacher's use only | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Question No | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
| Marks | | | | | | | | | | | |
| Question No | 11 | | | | | | | | | | |
| Marks | | | | | | | | | | | |

## Question No: 1 ( Marks: 10 )

Children Hospital Lahore has monthly demand of 6000 units of a tablet named Flagyl. Each tablet cost Rs. 6 and the cost of one purchase is Rs. 950.
Keeping that stock for a year in the hospitals cost Rs. 3 per unit. The shortage cost for a

year is Rs. 10 per unit, Determine the following

(i). The Optimum Order Quantity
(ii). The time between the orders
(iii). The number of orders per year
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(v). The time of orders being held

Question No: 2 ( Marks: 10 )

Solve the following problem with Big M – Method.

$$Minimize. Z = 3X_1 + 2X_2 + X_3$$

*Subject to.*

$$X_1 + X_2 = 7$$
$$3X_1 + X_2 + X_3 \geq 10$$
$$X_1,\ X_2,\ X_3 \geq 0$$

Question No: 3 ( Marks: 10 )

Write the DUAL of the following primal problem.

$$Maximize. Z = -5X_1 + 2X_2$$

*Subject to.*

$$-X_1 + X_2 \leq -2$$
$$2X_1 + 3X_2 \leq 5$$
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Question No: 4 ( Marks: 10 )

Solve the transportation model starting with the Vogal's Approximation Method.

| | 1 | 2 | 3 | 4 | **Supply** |
|---|---|---|---|---|---|
| 1 | 10 | 2 | 20 | 11 | **15** |
| 2 | 12 | 7 | 9 | 20 | **25** |

| | | | | | |
|---|---|---|---|---|---|
| 3 | 4 | 14 | 16 | 18 | **10** |
| **Demand** | **5** | **15** | **15** | **15** | |

Question No: 5 ( Marks: 10 )

In a college, every 15 minutes one student arrives for admission information. The staff in the only information counter takes 10 minutes for serving a student on an average. State suitable assumption and find.

I. The average queue length.
II. Increase in the arrival rate in order to justify for second counter (when the waiting time of a student is atleast 15 minutes the management will increase one more counter).

Question No: 6 ( Marks: 10 )

A machine costs Rs. 8000. Annual operating cost are Rs. 1000 for the first year, and then increase by Rs. 500 every year. Resale price are Rs. 4000 for the first year and then decrease by Rs. 500 every year. Determine at which age it is profitable to replace the machine.

Question No: 7 ( Marks: 1 ) - Please choose one

An unrestricted primal variable will result in an equality dual constraint.

- True
- False

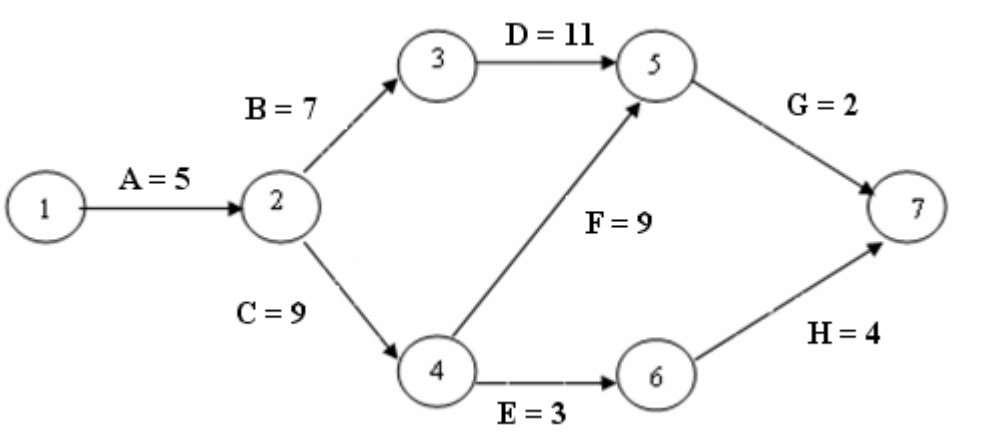
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In purchasing model with shortage, the formula of order size is

- $Q^* = \sqrt{\frac{2C_2D}{C_3}}$
- $Q^* = \sqrt{\frac{2C_2D}{C_3}} \quad \sqrt{\frac{C_3+C_4}{C_4}}$
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- $Q^* = \sqrt{\frac{2C_2D}{C_3(1-D/R)}} \quad \sqrt{\frac{C_3+C_4}{C_4}}$

Question No: 9 ( Marks: 1 ) - Please choose one




In Above Network Diagram the critical path is

- (a) A – B – D – G
- (b) A – C – E – H
- (c) A – C – F – G
- (d) Both (a) and (c)

Question No: 10 ( Marks: 1 ) - Please choose one

In assignment problem each job requires

- Exactly one resource
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- At most one resource

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The dual of the dual problem yields the original primal

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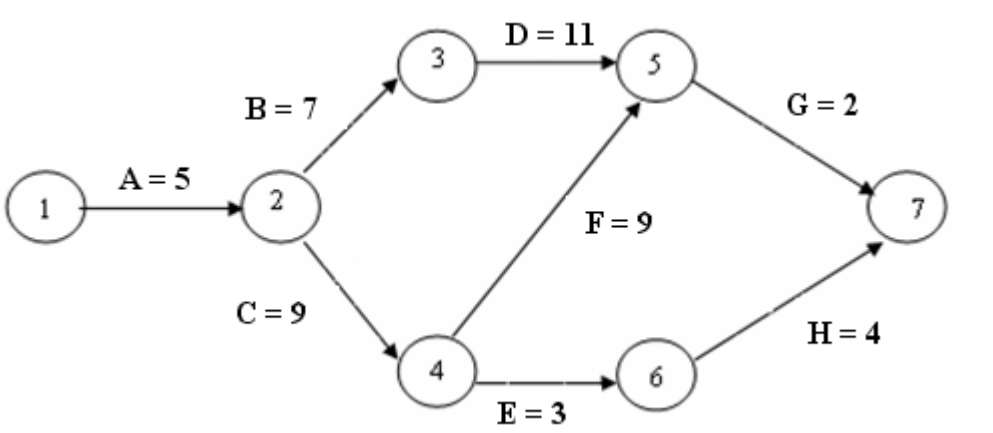
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- True
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# MTH601 Operations Research

## Final Term Examination - February 2005

**Time Allowed: 150 Minutes**

## Instructions

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**3.** In Q3 you can either draw the network diagram or simply write the asked terms for each activity in the exam software, whichever you think is easy for you. But remember that you have to show your work on exam application correspond to each question.

**Total Marks: 50** **Total Questions: 8**

**Question No. 1** **Marks : 4**

Define the followings.

(a) Impossible Assignment
(b) Unbounded Solution.

**Question No. 2** **Marks : 10**

Consider the problem

$$\text{maximize } z = x_1 + 5x_2 + 3x_3$$

Subject to

$$x_1 + 2x_2 + x_3 = 3$$
$$2x_1 - x_2 = 4$$
$$x_1, x_2, x_3 \geq 0$$

The variable $x_3$ plays the role of a slack. Use *big M-method* to solve the problem.

**Question No. 3** **Marks : 4**

How the problem of the degeneracy arises in a transportation problem?

**Question No. 4** **Marks : 4**

Define the followings in Queuing Theory.

a. Customer
b. Server
**c.** Priority

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What are the variants of the simplex method? Write few lines about the followings.

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Five different machines can process any of the five required jobs with different profits resulting from each assignment.

Find the maximum profit possible through optimum assignments.

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How can the linear programming be applied to management problem?

**Question No. 8** **Marks : 10**

Consider the transportation problem having the following parameter table.

Use the northwest corner rule to obtain an initial basic feasible solution.

# MTH601 - Operations Research - Final Term Paper

Fall 2005 – session 6

http://www.vuzs.net/study-portals/bscs-study-portal.html

Time Allowed: 150 Minutes

Total Marks: 50 Total Questions: 17

**MTH601 - Operations Research - Q. No. 1 M - 01**

Minimizing $Z = \sum_{i=1}^{n} c_i x_i$ is equivalent to

- 
- 
- 
- None of the above

**MTH601 - Operations Research - Q. No. 2 M - 01**

**According to_____ strategy, replace an item if it fails before the optimum period 'P'**

- IR
- IPR
- CPR (Common Preventive Replacement)

**MTH601 - Operations Research - Q. No. 3 M – 03**

A manufacturer has distribution centers at X, Y and Z. These centers have availability of 45, 20 and 30 units of the product, His retail outlets at A, B, C, D and E require 25, 10, 20, 30 and 15 units respectively. The transportation cost per unit between each centre and each outlet is given in the table below.

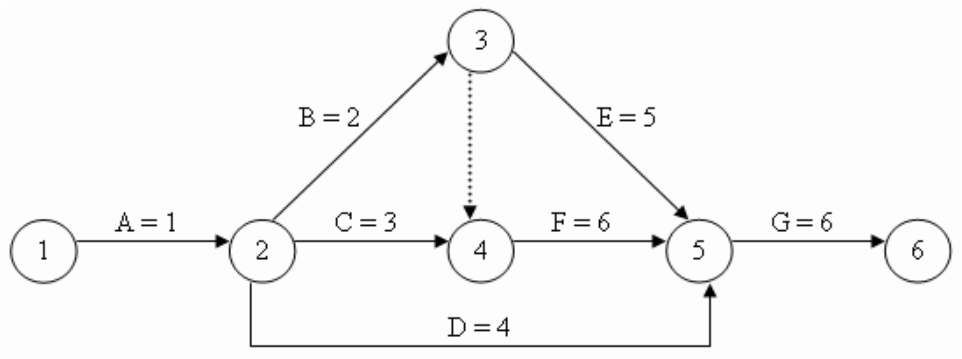
| Distribution Centers | Retail Outlets | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | E |
| X | 50 | 30 | 40 | 50 | 50 |
| Y | 35 | 30 | 100 | 45 | 60 |
| Z | 40 | 60 | 85 | 35 | 30 |

Is this model balanced? If "NO" balance the above Transportation model.

http://www.vuzs.net/study-portals/bscs-study-portal.html

**MTH601 - Operations Research - Q. No. 4 M – 05**

**Compute Latest start time (LST) and Latest Finish Time LFT for each activity of the following network, if the project is to be completed by the deadline of day 16.**




**MTH601 - Operations Research - Q. No. 5 M - 01**
In purchasing model with shortage, the formula of Ordering Size is

**MTH601 - Operations Research - Q. No. 6 M - 01**
Similarity between CPM and PERT is

- Both are used to plan the scheduling of individual activities that make up a project
- Both can be used to determine the earliest/latest stat and finish time for each activity
- All of the above

**MTH601 - Operations Research - Q. No. 7 M - 05**

The arrival rate of customers at a banking counter follows Poisson distribution with a mean of 45 customers per hour. The service rate of the counter clerk also follows Poisson distribution with a mean of 60 customers per hour.
Find the following
**1.** The probability of having 10 customers in the system
**2.** Average number of customers waiting in the system
**3.** Average number of customers waiting in the queue
**4.** Average waiting time of customers in the system
**5.** Average waiting time of customers in the queue

**MTH601 - Operations Research - Q. No. 8 M - 01**
The formula for expected number of customers in a system is

- $L = \lambda /(\mu - \lambda)$
- $L = \lambda /(\lambda - \mu)$ **o** $L = \mu /(\lambda - \mu)$ **o** $L = \mu /(\mu - \lambda)$

**MTH601 - Operations Research - Q. No. 9 M - 01**
For LP problem with '*n*' decision variables, each of its corner point solution is at the intersection of constraint boundaries.

- $(n - 1)$ constraint boundaries
- *'n'* constraint boundaries
- $(n + 1)$ constraint boundaries
- None of the above

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**MTH601 - Operations Research - Q. No. 10 M - 01**

In two dimensional LP problem we need only two lines to identify an extreme point. If in any two dimensional problem there are three or more intersecting lines at the corner point then this indicate that

- Problem has Degenerate Solution
- One of the constraints is redundant
- Both First and Second
- None of the above

**MTH601 - Operations Research - Q. No. 11 M – 10**

Consider the following LP problem
Minimize Z= 12 x1 + 18 x2 + 15 x3
Subject to

$4\ x1 + 8\ x2 + 6\ x3 \geq 64$
$3\ x1 + 6\ x2 + 12\ x3 \geq 96$
$x1\ ,\ x2\ ,\ x3 \geq 0$

(a). write the associated Dual Problem of the above Primal Problem
(b). The optimal table for phase-I of the above model is given below

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $R_1$ | $R_2$ | R.H.S |
|---|---|---|---|---|---|---|---|---|
| Z | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| $x_2$ | 1/2 | 1 | 0 | -1/5 | 1/10 | 1/5 | -1/10 | 16/5 |
| $x_3$ | 0 | 0 | 1 | 1/10 | -2/15 | -1/10 | 2/15 | 32/5 |

Using this above table find the Optimal Solution for Phase-II

**MTH601 - Operations Research - Q. No. 12 M - 01**

A transportation problem is degenerate, if while deriving a feasible solution an allocation to any cell

- Satisfies the Column requirements
- Satisfies the Row requirements
- Satisfies column as well as row requirements simultaneously
- None of the above

**MTH601 - Operations Research - Q. No. 13 M – 01**

Consider the following simplex table corresponding to maximization problem.

| Basic | $x_1$ | $x_2$ | $x_3$ | $S_1$ | $S_2$ | $S_3$ | R.H.S |
|---|---|---|---|---|---|---|---|
| Z | -4 | -3 | -2 | 0 | 0 | 0 | 0 |
| $x_1$ | 0 | 0 | 1 | 1 | 0 | 0 | 8 |
| $S_2$ | -2 | 1 | 3 | 0 | 1 | 0 | 9 |
| $S_3$ | 0 | 2 | 1 | 0 | 0 | 1 | 3 |

This is the indication that

- Problem has Multiple Optimal Solutions
- Problem has Unbounded Solutions
- Problem has Infeasible Solutions
- None of the above

**MTH601 - Operations Research - Q. No. 14 M – 07**

A manager has to allocate five different jobs to five workers. Depending on the efficiency and the capacity of the individual, the times taken by each worker to complete a job is given in the following table.

| Jobs | Workers | | | | |
|---|---|---|---|---|---|
| | W1 | W2 | W3 | W4 | W5 |
| 1 | 10 | 12 | 15 | 12 | 8 |
| 2 | 7 | 16 | 14 | 14 | 11 |
| 3 | 13 | 14 | 7 | 9 | 9 |
| 4 | 12 | 10 | 11 | 13 | 10 |
| 5 | 8 | 13 | 15 | 11 | 15 |

How the manager should assigned one job to a worker so as to minimize the total man-hours?

**MTH601 - Operations Research - Q. No. 15 M – 05**

Consider the following LP problem

$$\text{Maximize } Z = 10x_1 + 3x_2$$

Subject to

$$2x_1 + 3x_2 \leq 18$$

$$6x_1 + 5x_2 \geq 60$$

$$x_1, x_2 \geq 0$$

Solve the above problem by Graphical Method.

http://www.vuzs.net/study-portals/bscs-study-portal.html

**MTH601 - Operations Research - Q. No. 16 M - 01**

The Phase-I objective function Minimize $R_0 = \sum_{i=1}^{n} R_i$ , where $R_i$ are artificial variables is obtained by

- Dividing the Big-M method's objective function by 'M' and then dropping the negligible terms
- Dividing the Big-M method's objective function by '–M' and then dropping the negligible terms
- Multiply the Big-M method's objective function by 'M' and then dropping the negligible terms
- Multiply the Big-M method's objective function by '–M' and then dropping the negligible terms

**MTH601 - Operations Research - Q. No. 17 M – 05**

The following table is an initial table of M-technique for maximization problem.

| Basic | $x_1$ | $x_2$ | $x_3$ | $S_1$ | $S_2$ | R | R.H.S |
|---|---|---|---|---|---|---|---|
| Z | -3M-3 | -2-4M | 2M-3 | 0 | M | 0 | -8M |
| $S_1$ | 2 | 1 | 1 | 1 | 0 | 0 | 2 |
| R | 3 | 4 | 2 | 0 | -1 | 1 | 8 |

Write the original Mathematical Model (not the standard form of the model) from the above given table.

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